



PHD RESEARCH · XAI IN MEDICINE

An Interpretable Dual Graph Autoencoder Framework

for Autism Spectrum Disorder Detection from Resting-State fMRI

Georgia Kapetadimitri

Department of Applied Informatics · University of Macedonia

Supervisor: Eftychios Protopapadakis

Why Interpretable ASD Detection?

Autism Spectrum Disorder (ASD)

is a heterogeneous neurodevelopmental condition with a strong neural-circuit basis. Resting-state fMRI captures intrinsic functional connectivity — a natural fit for graph-based representation, brain regions as nodes, their interactions as edges.

The clinical bottleneck

Diagnosis remains behaviour-based and time-intensive. Imaging-based tools could complement clinical assessment — but only if their decisions are transparent enough for clinicians to trust.

1 in 31

U.S. children diagnosed with ASD (CDC 2025)

Black-box

Most existing fMRI classifiers offer no clinical insight

Networks

ASD increasingly understood as a connectivity disorder

The Interpretability Gap in fMRI-based ASD Classifiers



Existing approaches

- Post-hoc attribution via GNNExplainer, Grad-CAM, etc.
- Explanations are unstable, model-specific, and rarely tied to neurobiology



Our research question

Can we build a classifier whose predictive features are themselves edge-level explanations — not bolted on after training?

- Intrinsically interpretable
- Class-specific generative modelling of brain graphs
- Edge-level features grounded in functional connectivity

Contribution

A dual graph autoencoder framework that produces clinically interpretable features by construction.



Two class-specific GAEs

Separate generative models for ASD and Control populations, each specialised through a margin-based loss.



Per-edge ΔE features

Reconstruction-error differences across 6,105 edges – each tied to one anatomical connection. Top 300 are selected for classification.



Neurobiological alignment

Top-ranked edges concentrate on regions with established ASD literature support.



Statistically validated

9-fold stratified CV with permutation testing ($p = 0.001$) confirms above-chance signal.

Dataset: ABIDE

Autism Brain Imaging Data Exchange — multi-site resting-state fMRI consortium.

862

subjects (after QC)

403

ASD

459

Control

111

ROIs (Harvard–Oxford atlas)



Preprocessing pipeline

- C-PAC pipeline · motion correction, slice timing, nuisance regression
- 111 ROIs defined by the Harvard–Oxford atlas (cortical + subcortical)
- Mean BOLD time-series computed within each ROI
- Multi-site heterogeneity: 17 acquisition sites — a known difficulty driver on ABIDE

Building the Brain Graph

Nodes V

111 (cortical + subcortical) ROIs from the Harvard–Oxford atlas.

Node features X

Each node's features are its Pearson correlations with all 111 ROIs (a 111-dim vector per node).

Edges E

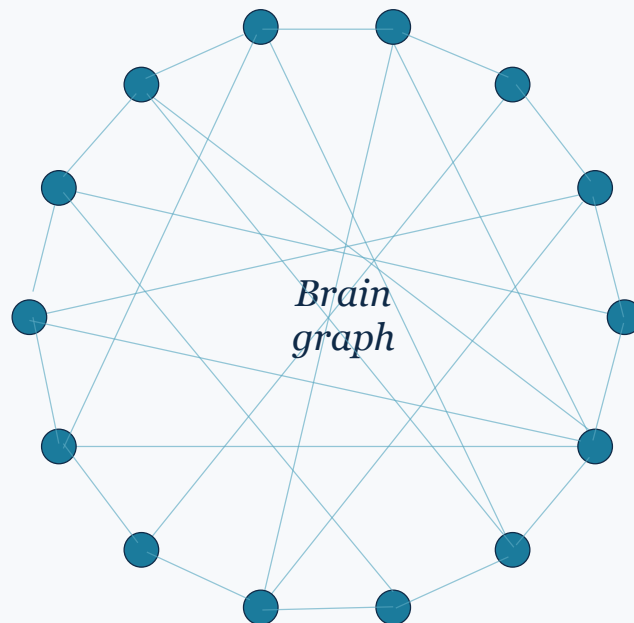
Fully connected graph — every region pair forms an edge.

Edge weights w

Partial correlations between BOLD time series — the partial functional connectivity (FC) matrix $A \in \mathbb{R}^{111 \times 111}$

Total edges

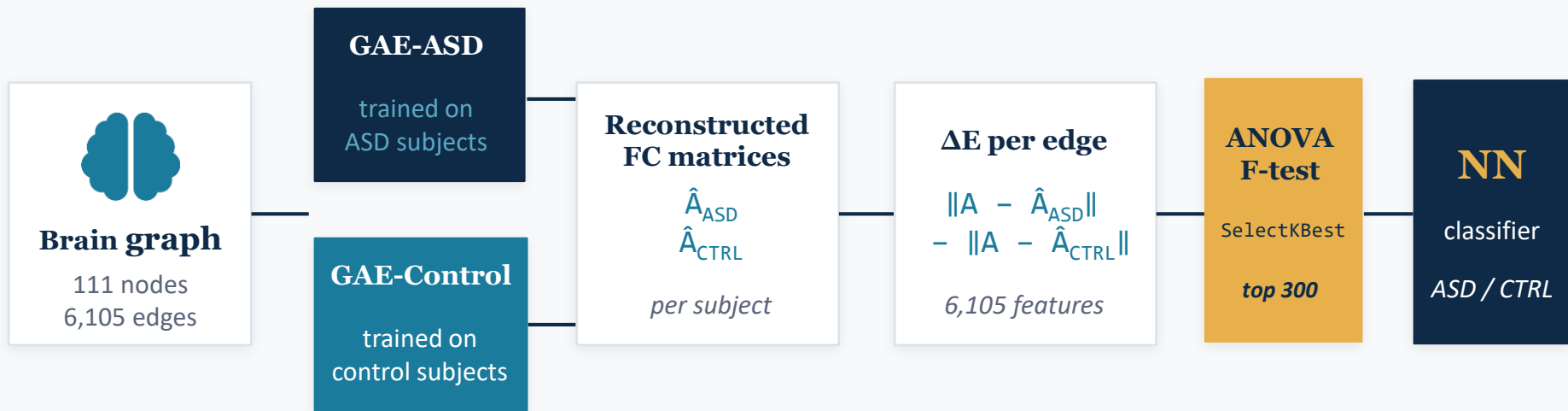
6,105 (upper triangle of A matrix) unique edges per subject



Per-subject · 111 nodes · 6,105 edges

Framework Overview

Two GAEs trained separately on each class — reconstruction errors become interpretable features.



Each subject's FC matrix is reconstructed by both GAEs. The per-edge difference in reconstruction quality, ΔE , drives classification — and points directly to which connections distinguish ASD subjects from controls.

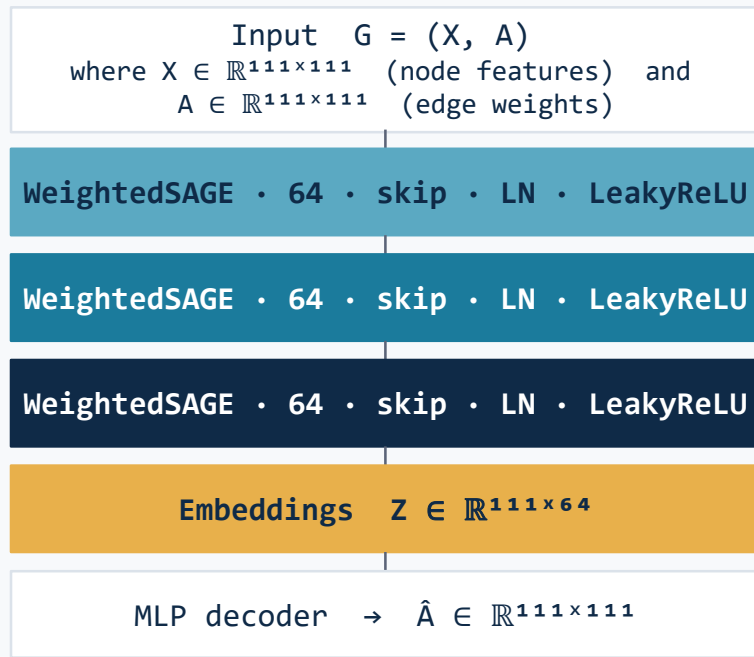
Encoder: 3-Layer GraphSAGE with Residual Connections

Why GraphSAGE

Inductive sampling-based aggregation; works on dense per-subject graphs with heterogeneous neighborhoods.

Decoder

MLP applied to each edge's concatenated node embeddings $[z_i || z_j]$
 → predicted edge weight, yielding the reconstructed FC matrix $\hat{A}$.



Margin-Based Specialization Loss

Force each GAE to reconstruct only its own class — and to fail on the opposite class.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}}(\text{own}) + \alpha \cdot \max(0, m - \mathcal{L}_{\text{recon}}(\text{opposite}))$$

the GAE is rewarded for reconstructing in-class graphs and penalised when it also reconstructs out-of-class ones — up to a margin m

$\mathcal{L}_{\text{recon}}$

Weighted MSE between A (partial-correlation adjacency) and $\hat{A}$.

$$m = 0.02, \alpha = 1$$

Margin and specialization weight — modest values that encourage class separation without destabilizing reconstruction

Augmentation

One perturbed copy per target-class graph during training (Gaussian noise $\sigma = 0.02$ added to edge weights).

Interpretable Feature: ΔE per Edge

For each subject, we compute the per-edge difference between the two GAEs' reconstruction errors:

$$\Delta E = \text{Err}_{\text{ASD}} - \text{Err}_{\text{CTL}}$$

$\Delta E > 0 \rightarrow$ GAE-ASD struggles $\rightarrow$ ASD-atypical edge

$\Delta E < 0 \rightarrow$ GAE-CTL struggles $\rightarrow$ control-atypical edge

Each subject is therefore represented by a 6,105-dimensional ΔE vector — every entry tied to a concrete anatomical edge. The top 300 are then selected by ANOVA F-test for classification.

Classification & Evaluation Protocol



Feature selection

ANOVA F-test (SelectKBest) retains the top 300 ΔE features (6,105 $\rightarrow$ 300). Selection — not transformation — so each retained feature remains a named anatomical edge.



Classifier

Lightweight feed-forward neural network operating on the 300 selected ΔE features.



Cross-validation

9-fold stratified CV — preserves the ASD/Control class balance per fold.



Stability analysis

Top edges identified across folds — guarantees that interpretable features are not artefacts of a single split.

Results

0.578

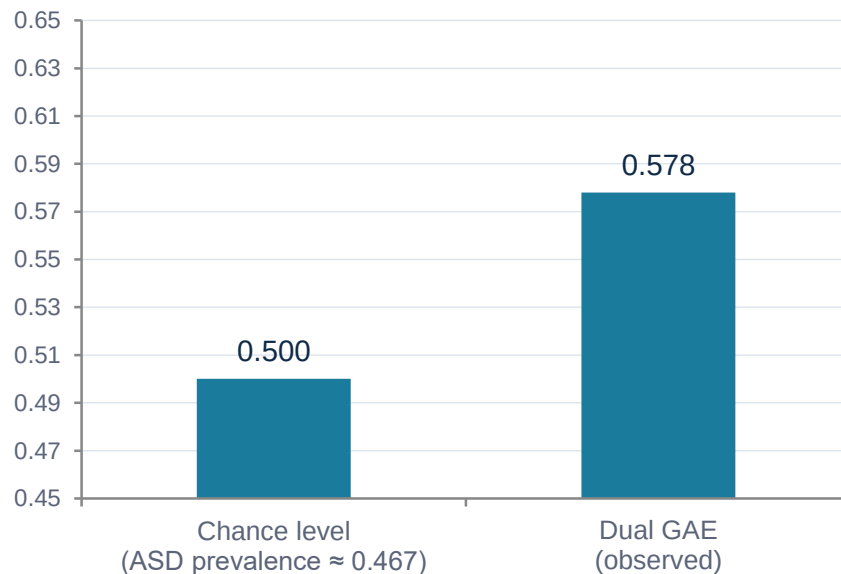
mean classification accuracy

9-fold stratified cross-validation

p = 0.001

permutation test (1,000 shuffles)

Mean accuracy (9-fold stratified CV)



What the Model Looks At

Three most consistently ranked ROIs across CV folds — all independently implicated in ASD by prior neuroimaging studies.

#1 Left inferior temporal gyrus

(temporo-occipital part, node 82)

PRIOR FINDING

Atypical gray matter properties reported in both low- and high-functioning ASD populations [Cai et al., 2018]

#2 Right angular gyrus

(node 34)

PRIOR FINDING

Altered functional connectivity in ASD relative to controls [Xu et al., 2019]

#3 Left superior parietal lobule

(node 80)

PRIOR FINDING

Differential activation in ASD during spatial and attentional tasks — broader involvement of parietal circuitry in ASD-related cognitive processes [Hamilton, 2013]

Discussion & Limitations



What the result means

- Modest accuracy is consistent with multi-site ABIDE difficulty (typically 55–70%).
- Permutation $p = 0.001$ confirms the signal is real, not noise.
- Critically: ΔE features are interpretable by construction — the same vector that drives prediction also localises atypical connectivity.
- Identified regions overlap with prior fMRI ASD literature — biological plausibility, not just statistical artefacts.



Limitations

- Site-related heterogeneity in ABIDE caps achievable accuracy.
- Atlas dependence — top-ranked regions are specific to the Harvard–Oxford parcellation.
- Partial correlations are sensitive to scan length and preprocessing choices.

Conclusion

Takeaway

Instead of training a black-box classifier and explaining it after the fact, we built a model whose features are the explanation — every classification decision traces back to specific anatomical edges.